Managing Big Data Group Project

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Executive Summary

This project designs a full enterprise data architecture for Walmart and introduces **GreenCart Advisor**, a sustainability-focused recommendation system. The architecture integrates **MySQL** for reliable master data, **MongoDB** for large and dynamic operational data, and **Hive on Hadoop** for scalable analytics. Using real and synthetic data, the system replicates Walmart's enterprise environment.

GreenCart Advisor recommends eco-friendly product alternatives in real time and records customer responses, generating new behavioral datasets such as willingness-to-pay for sustainability and acceptance of greener options—data Walmart does not currently collect. A dedicated **GreenCart star-schema warehouse** supports ESG analytics, customer segmentation, and long-term sustainability insights. Overall, the solution demonstrates how Walmart can combine operational efficiency with data-driven sustainability decision-making.

1. Existing Systems

1.1 Data Needs for Operations and Decision-Making

Walmart's existing systems must support high-volume operations such as checkout, promotions, payments, and inventory, while also enabling long-term strategic analysis. These needs fall into three categories: **OLTP transactions**, **flexible operational intelligence**, and **large-scale analytics**. MySQL, MongoDB, and Hive respectively fulfill these requirements.

1.2 OLTP System (MySQL)

MySQL stores core, high-integrity entities such as customers, products, stores, inventory, and payment methods. These tables are normalized with strict relational constraints to maintain accuracy in identity management, product catalog updates, and inventory control. Continuous updates and the need for ACID guarantees make MySQL the appropriate choice for this layer.

1.3 NoSQL System (MongoDB)

MongoDB manages large, fast-changing datasets including promotions, loyalty attributes, and detailed transaction logs. Its flexible document model supports rapid schema updates and allows retrieval of entire customer histories in a single query. Horizontal scalability and high availability enable MongoDB to support Walmart's peak operational loads while remaining decoupled from rigid relational structures.

1.4 Hive Big-Data Warehouse

Large operational datasets are periodically exported from MongoDB into Hive via S3 and HDFS. Hive external tables enable efficient analytical queries across multi-year data, supporting revenue analysis, customer segmentation, store performance tracking, and promotion effectiveness. Hive's distributed processing ensures reliability at the scale required by Walmart.

1.5 Implementation Challenges

Key challenges included maintaining schema consistency across systems, performing large-scale data movement, correcting SerDe configurations for Hive ingestion, and scaling EMR clusters to join multi-million records efficiently. Iterative ETL validation ensured accuracy and performance across all data layers.

2. New Business Application: GreenCart Advisor

2.1 Motivation

Although Walmart emphasizes sustainability publicly, customers lack real-time information on the environmental impact of their purchases. GreenCart Advisor fills this gap by recommending greener alternatives and recording how customers respond. This creates a new behavioral dataset—*real-time willingness-to-pay for sustainability*—that is not captured by Walmart today.

2.2 Dataset Design

GreenCart introduces two major data groups integrated into the OLTP environment:

1. **Sustainability Profiles:** Product sustainability scores, carbon emissions, resource usage, packaging, and customer sustainability goals.
2. **Recommendation Outputs:** Logged suggestions, price differences, sustainability gains, and acceptance or rejection.

These datasets enable fine-grained sustainability analytics and behavioral modeling.

2.3 Front-End Implementation

The prototype UI, built with Streamlit, allows customers to browse products, receive sustainable alternatives, and accept or reject suggestions. Each interaction writes new data to MySQL. The interface computes sustainability gains in real time and demonstrates live database connectivity and persistent behavioral logging.

2.4 Business Value

GreenCart delivers value through behavioral nudging, improved inventory management for sustainable items, and richer analytics on price sensitivity and environmental impact. It equips Walmart with actionable insights to enhance its ESG strategy, product development, and customer experience.

3. Decision-Making and Insights

3.1 Purpose of the GreenCart Data Warehouse

Operational systems are optimized for transactions, not analytics. To support advanced sustainability analysis, a **dedicated GreenCart data warehouse** was built using a star schema. The warehouse improves performance, isolates analytical workloads, and centralizes all sustainability-related information.

3.2 Star Schema Design

Dimensions include Customer, Date, Product, Loyalty Tier, Sustainability Goals, and Channel. The fact table—**FactGreenRecommendation**—stores each recommendation event with sustainability gains, price differences, and acceptance outcomes. This design supports KPI reporting, elasticity modeling, and multi-dimensional analysis.

3.3 ETL Pipeline

The ETL process uses Python to simulate customer behaviors and seed realistic recommendation logs. Transformations include surrogate key assignment, standardized sustainability metrics, and dimensional mapping. The final warehouse tables support clean, analytical-ready reporting.

3.4 Key Analytical Insights

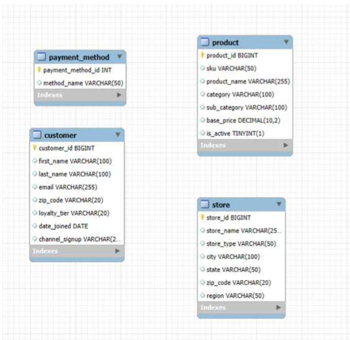
Customers are willing to choose sustainable alternatives when the price increase is small (under $1.50), but acceptance declines sharply for premiums above $5. Higher loyalty-tier customers are more likely to adopt these greener options. Walmart also faces strategic and reputational risk in high-revenue categories with low sustainability scores. Finally, repeated sustainable swaps in common grocery items can generate significant cumulative carbon reduction across the customer base.

3.5 Strategic Impact

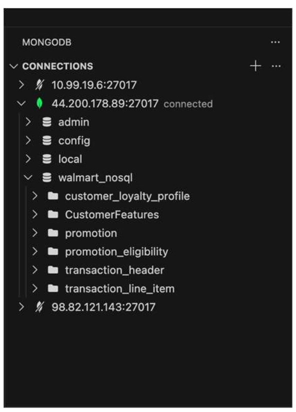
The GreenCart warehouse enables Walmart to quantify sustainability behavior, support ESG reporting, identify environmentally risky product categories, and personalize marketing for eco-conscious customers. It transforms sustainability from a qualitative initiative into a measurable, data-driven business capability.

PART 1

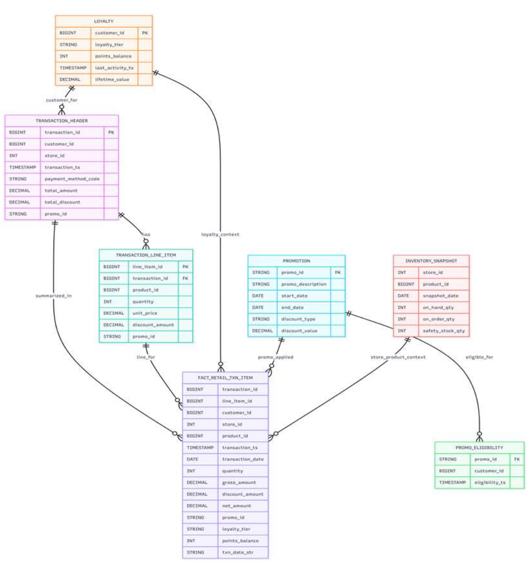
MySQL Database:



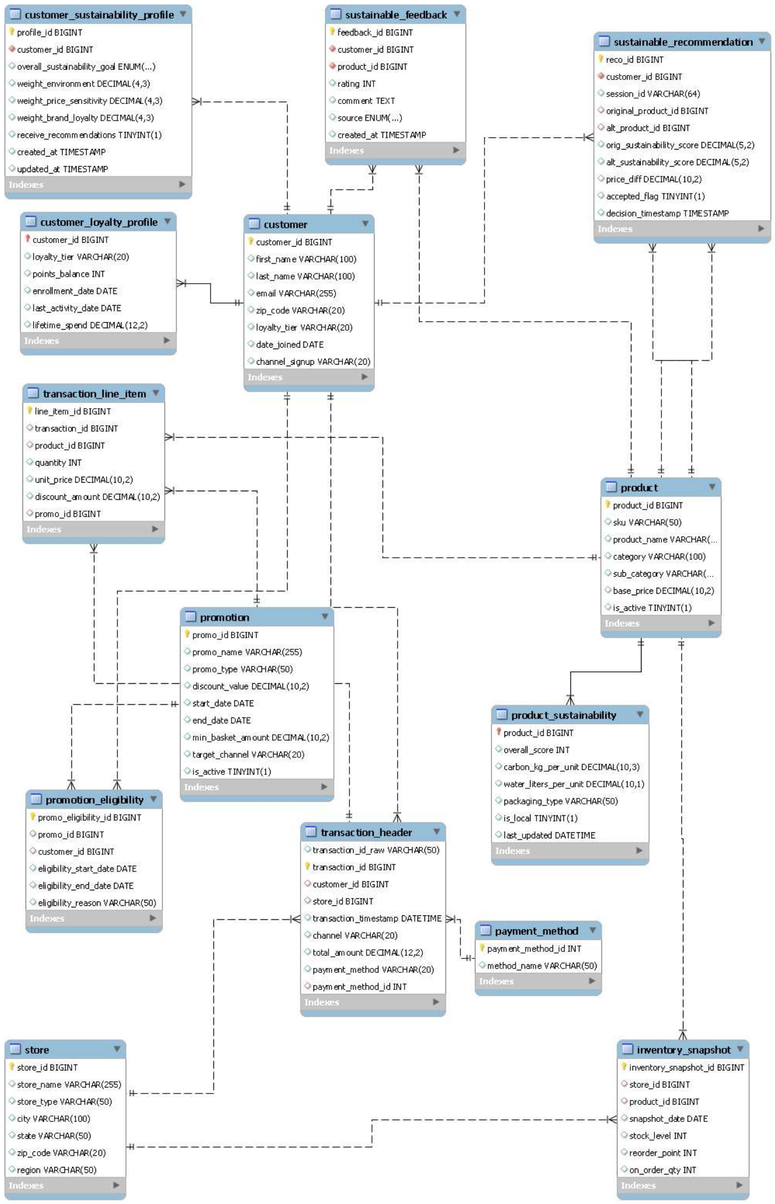
Database in MongoDB



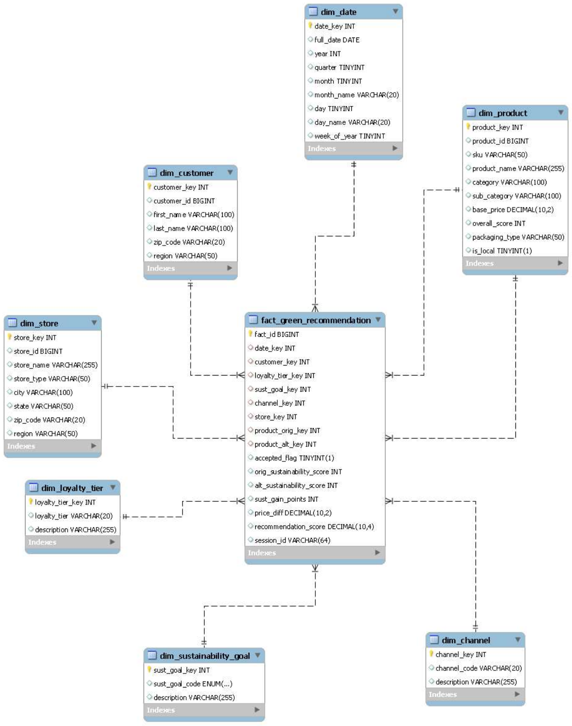
Database in Hive:



PART 2 ER Diagram

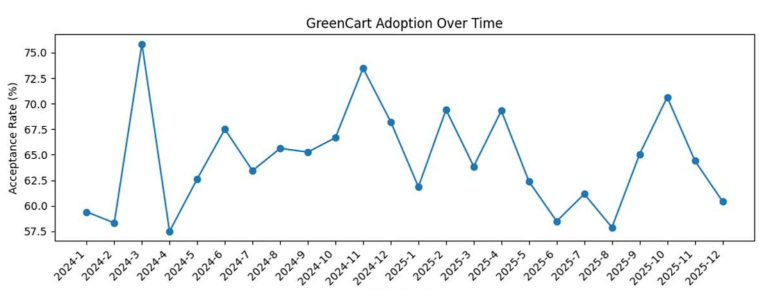


PART 3 ER Diagram



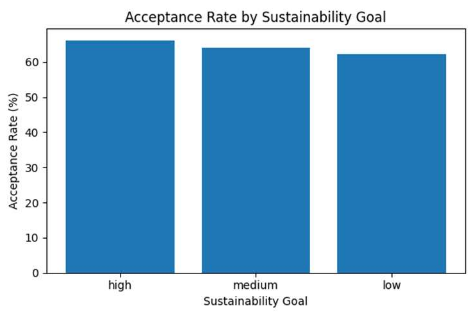
Part 3 Queries and Visualizations

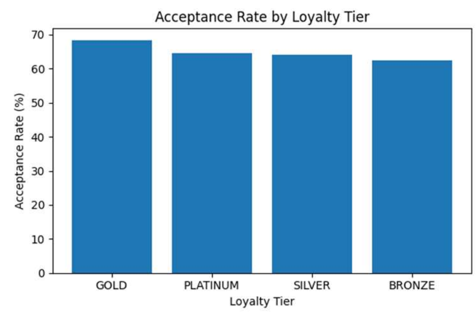
```
# --------------------------------
sql_q4 = """
SELECT
    dd.year,
    dd.month,
    COUNT(*) AS total_recos,
    SUM(f.accepted_flag) AS accepted_recos,
    ROUND(SUM(f.accepted_flag)/COUNT(*) * 100, 2) AS acceptance_rate_pct
FROM fact_green_recommendation f
JOIN dim_date dd
  ON f.date_key = dd.date_key
GROUP BY dd.year, dd.month
ORDER BY dd.year, dd.month;
"""
```



```
# --------------------------------
sql_q5 = """
SELECT
    dsg.sust_goal_code,
    COUNT(*) AS total_recos,
    SUM(f.accepted_flag) AS accepted,
    ROUND(SUM(f.accepted_flag)/COUNT(*) * 100, 2) AS acceptance_rate
FROM fact_green_recommendation f
JOIN dim_sustainability_goal dsg
  ON f.sust_goal_key = dsg.sust_goal_key
GROUP BY dsg.sust_goal_code
ORDER BY acceptance_rate DESC;
"""
```

```
# --------------------------------
sql_q1 = """
SELECT
    dlt.loyalty_tier,
    COUNT(*) AS total_recos,
    SUM(f.accepted_flag) AS accepted_recos,
    ROUND(SUM(f.accepted_flag)/COUNT(*) * 100, 2) AS acceptance_rate_pct
FROM fact_green_recommendation f
JOIN dim_loyalty_tier dlt
  ON f.loyalty_tier_key = dlt.loyalty_tier_key
GROUP BY dlt.loyalty_tier
ORDER BY acceptance_rate_pct DESC;
"""
```






```
# --------------------------------
sql_q12 = """
          SELECT dp.category, \
                 AVG(dp.overall_score) AS avg_sustainability_score, \
                 COUNT(*)              AS num_alt_products, \
                 AVG(dp.base_price)    AS avg_alt_price
          FROM fact_green_recommendation f
                   JOIN dim_product dp
                        ON f.product_alt_key = dp.product_key
          GROUP BY dp.category
          ORDER BY avg_sustainability_score DESC; \
          """
```

```
# --------------------------------
sql_q11 = """
          SELECT dlt.loyalty_tier, \
                 AVG(dp_alt.overall_score) AS avg_sustainability_score, \
                 COUNT(*)                  AS num_recommendations
          FROM fact_green_recommendation f
                   JOIN dim_loyalty_tier dlt
                        ON f.loyalty_tier_key = dlt.loyalty_tier_key
                   JOIN dim_product dp_alt
                        ON f.product_alt_key = dp_alt.product_key
          GROUP BY dlt.loyalty_tier
          ORDER BY avg_sustainability_score DESC; \
          """
```